

Part I

In this part we learn to use the MATLAB commands **norm** and **cond**.

MATLAB has a built-in command **norm** for the computation of vector norms and matrix norms. Create

$$\mathbf{x} = \begin{bmatrix} -2 \\ 3 \\ 5 \\ -1 \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} 1 & 0 & -2 & 5 \\ -3 & 2 & 1 & -4 \\ 5 & -6 & -1 & 0 \\ 2 & 5 & 0 & 12 \end{bmatrix}$$

Try the following commands:

>>norm(x,1)

This computes the 1-norm of **x**.

>>norm(x,2)

This computes the 2-norm of **x**.

>>norm(x)

The default norm for a vector is the 2-norm.

>>norm(x,inf)

This computes the ∞ -norm of **x**.

>>norm(A, 1)

This computes the 1-norm of **A**.

>>norm(A, 2)

This computes the 2-norm of **A**.

>>norm(A)

The default norm for a matrix is the 2-norm.

>>norm(A, inf)

This computes the ∞ -norm of **A**.

The condition number $\kappa(\mathbf{A})$ of a nonsingular $n \times n$ matrix **A** is defined by the formula

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \cdot \|\mathbf{A}^{-1}\|.$$

The three commonly used condition numbers are

$$\kappa_1(\mathbf{A}) = \|\mathbf{A}\|_1 \cdot \|\mathbf{A}^{-1}\|_1, \quad \kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2 \cdot \|\mathbf{A}^{-1}\|_2 \quad \text{and} \quad \kappa_\infty(\mathbf{A}) = \|\mathbf{A}\|_\infty \cdot \|\mathbf{A}^{-1}\|_\infty.$$

MATLAB has a built-in command **cond** for the computation of the condition number of a matrix.

```
>>cond(A)
```

The default condition number for a matrix is the condition number $\kappa_2(\mathbf{A})$.

Of course you can also compute $\kappa_2(\mathbf{A})$ by the following commands:

```
>>norm(A)*norm(inv(A))
```

The MATLAB command **inv(A)** computes the inverse of the matrix **A**.

You can also compute $\kappa_1(\mathbf{A})$ and $\kappa_\infty(\mathbf{A})$:

```
>>cond(A,1)
```

This computes the condition number $\kappa_1(\mathbf{A})$.

```
>>cond(A, inf)
```

This computes the condition number $\kappa_\infty(\mathbf{A})$.

Clear all variables before you work on Part II